

Ontology-Grounded Knowledge Graph Reasoning for Clinical LLM Systems: A SNOMED CT-Driven Approach to Retrieval-Augmented Generation

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Abstract

Retrieval-augmented generation (RAG) over clinical documents helps large language models (LLMs) cite evidence, but vector-only retrieval struggles with multi-hop clinical reasoning such as drug-interaction checks or contraindication chains. This paper reports on a system that combines a SNOMED CT-aligned clinical knowledge graph with dense vector retrieval for clinical question answering. The KG contains approximately 45,000 entities and 180,000 relations drawn from SNOMED CT, ICD-10, de-identified EHR metadata, and public clinical guidelines, aligned through UMLS Concept Unique Identifiers (CUIs). At query time, entities extracted from the question are mapped to KG nodes and a schema-aware Cypher template retrieves triples within 1–2 hops; these triples are serialized and fused with passages retrieved by PubMedBERT and FAISS (Facebook AI Similarity Search) before generation. On a set of 150 expert-judged clinical questions, the hybrid system improves expert-rated accuracy by 13.3 percentage points, reduces unsupported claims by 5.9 points, and substantially improves evidence traceability relative to a vector-only RAG baseline. We view our contribution as an empirical replication, in the healthcare domain and with a specific ontology-alignment approach, of a broader finding that is now converging across several recent studies: ontology-grounded retrieval is a useful substrate for LLM-based clinical QA.

Keywords

Knowledge Graphs, SNOMED CT, Ontology Alignment, Retrieval-Augmented Generation, Clinical Decision Support, UMLS

1. Introduction

Clinicians answering a question such as “Is Metformin safe for a patient with Type 2 Diabetes and Stage 4 CKD?” rely on structured relationships between entities: *treats*, *contraindicatedWith*, and *hasRiskFactor*. Clinical ontologies such as SNOMED CT and ICD-10 encode these relationships and have supported clinical decision support systems for years [1, 2]. Large language models (LLMs) capture a lot of textual medical knowledge but do not consistently perform the kind of entity-level, multi-hop lookups that ontologies make explicit, and this gap is a known source of hallucinated medical claims [3].

Retrieval-augmented generation (RAG) [4] addresses part of this problem by grounding LLM output in retrieved text passages. In clinical RAG, however, vector similarity alone does not reliably surface the specific contraindication or interaction edge needed to answer a question, because those edges often are not stated explicitly in any single retrieved passage [5]. Recent work has therefore begun to combine KGs with RAG in the medical domain [6, 7, 8, 9], and a broader line of work on ontology-grounded RAG has reported similar benefits outside healthcare [10].

This paper describes a specific instantiation of this direction. Our system (Figure 1) has three stages: (i) a clinical KG constructed from SNOMED CT, ICD-10, de-identified EHR metadata, and public guidelines, with entities aligned through UMLS CUIs; (ii) a dual-path retriever that runs a schema-aware Cypher template over matched KG entities in parallel with a PubMedBERT plus FAISS¹ vector retrieval path; and (iii) a cross-encoder that fuses the two result sets before the LLM generates a grounded answer. We evaluate on 150 clinical questions rated by three domain experts.

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¹FAISS: Facebook AI Similarity Search [19].

What this paper contributes, honestly stated.

1. A reproducible recipe for constructing a SNOMED CT/ICD-10-aligned clinical KG from a small set of public sources, using UMLS CUIs for cross-vocabulary alignment.
2. A schema-aware retrieval procedure in which the question’s extracted entities drive a parameterised Cypher template that is specialised to a fixed set of 13 curated clinical relation types. The schema and a sample subgraph are released publicly.²
3. An expert-rated evaluation that replicates, in a new clinical setting, the finding that ontology-grounded RAG improves accuracy, reduces hallucination, and improves traceability relative to vector-only RAG [6, 10].
4. A worked example and per-category breakdown showing where the structured path helps most.

We are not claiming to be the first to combine KGs with clinical RAG; MedGraphRAG [6], MEG [7], HyKGE [8], MedRAG [9], and OG-RAG [10] have all explored closely related designs. Our contribution is a small, explicit system that makes two design choices (UMLS-CUI alignment across sources and a fixed schema of 13 relations driving a parameterised Cypher template) and reports how those choices perform under expert review.

2. Related Work

KG-augmented RAG in medicine. MedGraphRAG [6] introduces a triple-graph construction that links user documents to credible medical sources and controlled vocabularies, together with a U-retrieval procedure for traversal. Our system is architecturally smaller and does not attempt the three-tier linkage of MedGraphRAG; instead we assume SNOMED CT/ICD-10 as fixed grounding and focus on UMLS-CUI-based alignment at ingestion. MEG [7] takes a different route: it trains a lightweight mapping network that injects KG embeddings into an LLM, rather than retrieving textual triples at inference time. Our approach is retrieval-time only and does not modify the LLM weights, which matters for HIPAA-constrained deployments that want to keep the base model fixed. HyKGE [8] uses an LLM-generated hypothesis to probe a medical KG; MedRAG [9] enhances RAG with KG-elicited reasoning over a healthcare copilot benchmark. Both report accuracy and reliability gains consistent with ours, though neither uses SNOMED CT as the primary alignment target.

Ontology-grounded RAG outside medicine. OG-RAG [10] argues explicitly for ontology grounding in RAG and constructs a hypergraph representation of documents, with each hyperedge carrying ontology-grounded factual clusters. Our framing in the healthcare domain is compatible with OG-RAG’s thesis; we treat our results as domain-specific evidence for that broader claim.

Graph RAG and structured retrieval. GraphRAG [11] performs community-based summarisation over an LLM-built graph; KAPING [12] verbalises KG triples for prompt augmentation in zero-shot KGQA. Our retrieval is closer to KAPING in that we verbalise triples as natural-language evidence, but we use an externally-built SNOMED CT-aligned graph rather than an LLM-built one.

Clinical ontologies and semantic infrastructure. The UMLS [1] integrates over 200 biomedical vocabularies through CUIs; SNOMED CT [2] provides a comprehensive description-logic-based clinical terminology. Our KG is serialised as RDF [13] and can be queried with SPARQL [14]; the Neo4j deployment we use supports Cypher translation. Ontology alignment in our system follows established practice [15]. The relational ML framing for KGs [16] motivates the choice of graph-structured representation, although we do not learn embeddings over the KG in this work.

²Schema, sample triples, and Cypher templates: <https://github.com/harmanJLD/snomed-kg-rag>.

Positioning. Given the above, the honest positioning of this paper is: an empirical replication, in the SNOMED CT-aligned clinical setting, of the now-repeatedly-reported finding that ontology-grounded retrieval helps clinical RAG. The specific contributions are the UMLS-CUI alignment recipe, the fixed 13-relation schema, and the expert evaluation.

3. System Overview

Figure 1 shows the full pipeline. At ingestion we construct the KG; at query time we run the dual-path retriever and generation.

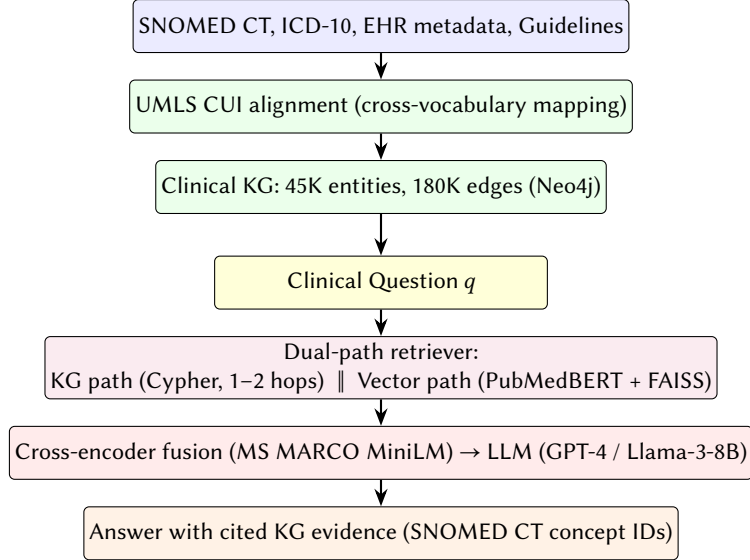


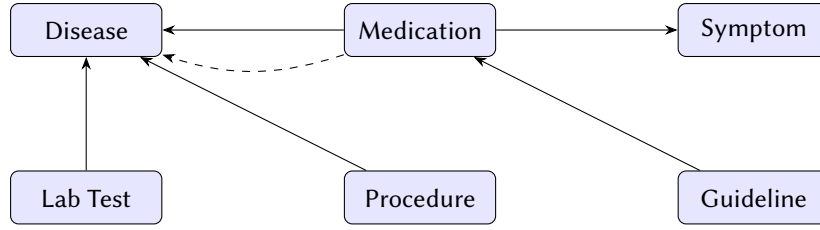
Figure 1: End-to-end system. Top: offline KG construction with UMLS CUI alignment. Bottom: online query processing with a schema-aware Cypher template on the KG path running in parallel with dense vector retrieval, fused before LLM generation.

3.1. KG schema

The KG schema (Figure 2) defines six core entity types (*Disease*, *Medication*, *Symptom*, *Lab Test*, *Procedure*, and *Guideline*), each typed by a SNOMED CT semantic category, and thirteen curated relation types: *treats*, *causes*, *contraindicatedWith*, *diagnosedBy*, *hasSideEffect*, *interactsWith*, *indicatedFor*, *partOf*, *findingSite*, *hasRiskFactor*, *monitoredBy*, *comorbidWith*, and *recommends* (used for guideline-to-medication and guideline-to-procedure edges). We chose a fixed schema (rather than an open LLM-generated schema as in some GraphRAG work) so that the Cypher template at query time can be specialised to these edge types; we return to this choice in §5.

3.2. KG construction

We populate the KG from three source types. From **SNOMED CT and ICD-10** we extract is-a hierarchies, finding-site relationships, and diagnostic category structure. From **de-identified EHR metadata** (synthetic note templates and schema-level field definitions used internally at the deployment site, with no patient-identifiable data and no individual record content; all templates were reviewed to ensure conformance with HIPAA Safe Harbor de-identification) we extract diagnosis/medication co-occurrence patterns, procedure/diagnosis links, and lab/condition correlations; clinical NER uses SciSpacy [17] and MedSpacy to extract mentions from these templates. From **public clinical guidelines** (CDC, WHO, AHA, ADA) we extract (guideline, recommends, medication) and (guideline, recommends, procedure) triples for a set of high-prevalence conditions.



Edges: Medication treats Disease; Medication hasSideEffect Symptom; Medication contraindicatedWith Disease (dashed); Lab Test diagnosedBy Disease; Procedure indicatedFor Disease; Guideline recommends Medication. Self-loops on Medication (interactsWith) and on Disease (causes, hasRiskFactor) are omitted for clarity. The full 13-relation schema is listed in §3.1.

Figure 2: KG schema: six core entity types with six representative edges from the thirteen-relation schema. All entities are typed by SNOMED CT semantic categories and identified by UMLS CUIs.

Alignment via UMLS CUIs. A recurring difficulty is that the same concept is expressed differently across sources: SNOMED CT code 44054006, ICD-10 code E11, and EHR abbreviation T2DM all refer to Type 2 Diabetes. We canonicalise every entity by its UMLS CUI (e.g., C0011860); exact CUI matches are used where available, and SciSpacy UMLS entity linking handles fuzzy text mentions. Entities that fail to resolve to a CUI above a confidence threshold are dropped; this is a coverage and precision trade-off we revisit in §5.

3.3. Query-time retrieval

Given question q :

KG path (symbolic). We run clinical NER on q , link extracted mentions to KG nodes via CUI normalisation, and execute a parameterised Cypher template. For a single matched entity the template retrieves directly-connected triples across the 13 relation types; when two or more entities are extracted we additionally retrieve paths of length 2 connecting them. Listing 1 shows the core 1-hop template. Retrieved triples are serialised as natural-language evidence with provenance tags, e.g., “Metformin [treats] Type 2 Diabetes (SNOMED: 44054006). Metformin [contraindicatedWith] Renal Failure (eGFR<30) (Source: ADA Guidelines 2024).”

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MATCH (n:Entity)-[r]->(m:Entity)
WHERE (n.cui IN $cuis OR m.cui IN $cuis)
AND type(r) IN $rels
RETURN n.label, type(r), m.label,
       n.cui, m.cui, r.source
LIMIT 50
  
```

Listing 1: Parameterised Cypher template used by the KG path. $\$cuis$ is the set of UMLS CUIs extracted from the question; $\$rels$ is the fixed schema of 13 curated relation types. The template matches edges where either endpoint is a linked entity, so triples in which the question entity appears as the object (e.g., guideline-to-medication recommendations) are also retrieved. The 2-hop extension is analogous.

Note that graph traversal here is performed by the Cypher engine, not by the graph itself; the KG is a queryable substrate for reasoning that happens externally.

Vector path (neural). q and the clinical document corpus are embedded with PubMedBERT [18]; top- k passages are retrieved from FAISS by cosine similarity.

Fusion. We concatenate the serialised KG triples and the retrieved passages, re-rank them with an MS MARCO MiniLM cross-encoder, and truncate to the LLM context window. The LLM prompt asks the model to answer q using *only* the provided context and to cite evidence by source identifier.

3.4. A worked example

To make the retrieval procedure concrete, we walk through one question from the evaluation set.

Question. “Can Metformin be safely prescribed to a 68-year-old with Type 2 Diabetes and Stage 4 chronic kidney disease (eGFR 22)?”

NER and CUI linking. SciSpacy extracts {Metformin, Type 2 Diabetes, Chronic Kidney Disease, eGFR}, linked to {C0025598, C0011860, C0022661, C0017654}.

Cypher result (1-hop and 2-hop). Figure 3 shows the subgraph returned. The template returns the triples: (Metformin, treats, T2DM), (T2DM, comorbidWith, CKD), (CKD, hasRiskFactor, RenalFailure), (Metformin, contraindicatedWith, RenalFailure), (ADA_Guideline, recommends, Metformin), (HbA1c, monitoredBy, T2DM).

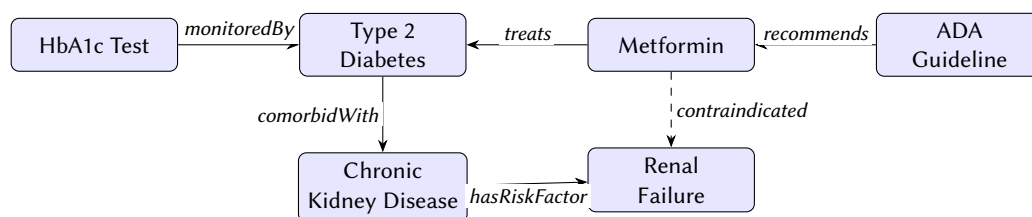


Figure 3: Subgraph actually returned for the worked example in §3.4. The Cypher template retrieved six triples at 1–2 hops from the linked entities. The path from Metformin through T2DM, CKD, Renal Failure, and back to Metformin via contraindicatedWith is the decisive chain for the answer.

Generated answer (abridged). “Metformin is the ADA-recommended first-line agent for Type 2 Diabetes [SNOMED:44054006, ADA 2024]. However, at eGFR 22 (Stage 4 CKD), Metformin is contraindicated due to increased lactic acidosis risk [SNOMED:91273001]. An alternative agent such as a DPP-4 inhibitor should be considered.”

The decisive edge for this answer, (Metformin, contraindicatedWith, RenalFailure), is present in the KG as a single typed relation but is not reliably surfaced by vector retrieval over clinical notes because the contraindication usually appears in prescribing guidelines rather than in patient-specific text.

4. Evaluation

4.1. Question set and provenance

We evaluate on 150 clinical questions spanning five categories: medication management (40 questions), differential diagnosis (30), procedure appropriateness (25), lab result interpretation (30), and guideline adherence (25). Questions were authored by the three domain experts (see below) based on (i) anonymised categories of information requests fielded by the UConn Student Health and Wellness Center IT team, and (ii) publicly available USMLE-style clinical case prompts. Each question has a reference answer agreed on by all three experts. We are aware the sample is modest in size and that questions authored by the rating experts can introduce anchoring; results should be read accordingly.

The expert panel is three people: two board-certified physicians (one internal medicine, one family medicine) and one clinical informaticist with >5 years’ EHR experience. Inter-rater agreement was measured with Krippendorff’s α on a 30-question calibration subset; $\alpha = 0.78$ for accuracy and $\alpha = 0.71$ for hallucination. Both fall in the tentative range under Krippendorff’s standard thresholds ($0.667 \leq \alpha < 0.80$ tentative; $\alpha \geq 0.80$ reliable) [20], so we treat the agreement as indicative rather than strong and discuss this further in §5.

4.2. Ethics and data availability

This study did not involve patient data or contact with patients. The KG was built from public ontologies (SNOMED CT, ICD-10), public clinical guidelines, and synthetic / schema-level note templates with no individual record content. The expert evaluation was conducted with three consenting adult collaborators who authored and rated the question set as part of their role. The study was reviewed by the UConn IRB and determined to be exempt under 45 CFR 46.104(d). The 150-question evaluation set, expert ratings, model outputs, and the schema and Cypher templates used in the experiments are released alongside the paper at <https://github.com/harmanJLD/snomed-kg-rag>; we omit the raw synthetic note templates because they encode internal site-specific field naming.

4.3. Systems compared

LLM-only: no retrieval; the model receives the question and a fixed instruction to “answer the clinical question and cite sources where possible.” **Standard RAG:** PubMedBERT embeddings and FAISS over the same clinical document corpus, no KG; the model receives the same instruction plus the top- k retrieved passages as context. **KG-RAG (ours):** dual-path retrieval as described in §3, with the same instruction. All three conditions use the same generation prompt template, the same max-context budget, and temperature 0.2; the only thing that varies is what is placed in the context window. We run all three configurations with two generation backends: GPT-4 (gpt-4-0613, API) and Llama-3-8B-Instruct (locally deployed on a single A100).

4.4. Metrics

Accuracy is expert-rated correctness on a 3-point scale (incorrect, partially correct, correct), reported as the fraction marked fully correct. **Hallucination rate** is the fraction of responses containing at least one factual claim that experts could not verify against an authoritative source. **Completeness** is the fraction of multi-part questions in which all aspects were addressed. **Interpretability (traceability)** is the fraction of answers for which the expert could identify a cited source for each load-bearing claim.

4.5. Results

Reading the numbers. KG-RAG improves accuracy over Standard RAG by 13.3 points with GPT-4 and 14.0 points with Llama-3-8B (Tables 1, 2). Hallucination drops by 5.9 points (GPT-4) and 10.7 points (Llama-3-8B). The largest gains in interpretability (+32 to +33 points) come from the cited SNOMED CT concept identifiers and guideline provenance the KG path carries through to the generated answer.

Table 1

Clinical QA results, GPT-4 as generator. Accuracy, Completeness, and Interpretability are higher-is-better; Hallucination is lower-is-better (↓). Δ vs. Standard RAG reported in percentage points.

Method	Acc.	Halluc. ↓	Compl.	Interp.
LLM-only	58.7%	34.2%	51.3%	22.0%
Standard RAG	71.3%	18.7%	68.0%	45.3%
KG-RAG (ours)	84.6%	12.8%	82.7%	78.0%
Δ vs. Std. RAG	+13.3	−5.9	+14.7	+32.7

Per category (Table 3), gains are largest on medication management (+17.2 points) and differential diagnosis (+15.1), the two categories most likely to require a contraindication or co-morbidity edge. The smallest gain is on lab interpretation (+5.8), where reference ranges and trend information already appear in retrieved text. Categories in Table 3 are ordered by gain magnitude.

Table 2

Clinical QA results, Llama-3-8B as generator.

Method	Acc.	Halluc. ↓	Compl.	Interp.
LLM-only	42.0%	47.3%	38.7%	14.7%
Standard RAG	59.3%	28.0%	56.0%	36.7%
KG-RAG (ours)	73.3%	17.3%	72.0%	68.7%
Δ vs. Std. RAG	+14.0	−10.7	+16.0	+32.0

Table 3Per-category accuracy gain of KG-RAG over Standard RAG (GPT-4, percentage points), ordered by Δ .

Category	Std. RAG	Δ KG-RAG
Medication management	66.5%	+17.2
Differential diagnosis	69.0%	+15.1
Guideline adherence	74.2%	+12.9
Procedure appropriateness	72.0%	+11.6
Lab result interpretation	75.1%	+5.8

5. Discussion

What we think actually drives the gain. The evidence from Table 3 and the worked example points to a specific mechanism: the KG path reliably surfaces single edges (contraindications, interactions, recommendations) that vector retrieval misses because those edges are not verbalised near the query terms in the document corpus. We do not interpret our numbers as evidence that KGs “reason” in place of the LLM; the reasoning still happens in the LLM. What the KG provides is a more predictable retrieval surface for exactly the edge types the schema enumerates.

Dependence on schema coverage. The flip side of the schema-aware Cypher template is that it works only for questions whose answer hinges on one of the 13 curated relation types. Questions that would need, say, a temporal or dose-response relation not in the schema gain nothing from the KG path. In our set this showed up as flat performance on some lab interpretation questions. A more expressive schema would probably help but also complicates the Cypher template.

Effect of LLM capability. The relative improvement is larger for Llama-3-8B than GPT-4, which is consistent with an interpretation where the KG compensates for weaker parametric knowledge. With a stronger generator (GPT-4), some of the KG-path information is already known to the model, and the KG’s contribution is concentrated on the narrower set of edges (specific contraindications, less common interactions) that even strong LLMs get wrong. We expect future, stronger LLMs to shrink the average accuracy gap while keeping the interpretability gap roughly constant, because the latter comes from *citing* sources, not from *knowing* them.

Relation to prior work. Our numbers are in the same range as those reported by MedGraphRAG [6] and MedRAG [9] on their respective benchmarks, and they reinforce the broader claim of OG-RAG [10] that ontology grounding is a useful RAG substrate. We do not claim our system is better; we claim the pattern generalises to a SNOMED CT/ICD-10 setting under expert review.

Limitations.

- *Sample size and inter-rater agreement.* N=150 with three raters limits statistical power. Both α values (0.78 for accuracy, 0.71 for hallucination) fall in the tentative range, not the reliable range, under standard Krippendorff thresholds.

- *Question authorship.* Questions were authored by the same experts who rated the outputs, which can introduce anchoring; a blinded external question set would be stronger.
- *Schema coverage.* 13 relation types is a deliberate simplification. The schema does not cover temporal relations (e.g., *precedes*, *follows*), dose-response or dose-dependent contraindications, drug-drug interaction severity grading, or patient-attribute-conditioned edges (age, pregnancy status). Questions whose answer hinges on any of these gain nothing from the KG path.
- *EHR-derived edges.* Co-occurrence-based edges from the de-identified template data are noisier than curated ontology edges; we do not yet quantify the contribution of each source separately.
- *Static KG.* Clinical guidelines change; our KG was built at a single point in time and we do not evaluate drift.
- *Single-site, single LLM-version.* All expert raters are affiliated with one institution, and the GPT-4 results reflect a single API snapshot (gpt-4-0613); subsequent model releases may shift the gap.

Future work. (i) An ablation by source (SNOMED CT only vs. +ICD-10 vs. +EHR vs. +Guidelines) to quantify each source’s marginal contribution. (ii) Automatic guideline ingestion so the KG can be kept current. (iii) A blinded external question set. (iv) Comparison against MedGraphRAG and OG-RAG on a shared benchmark. (v) Re-running the experiment with newer LLM releases (GPT-4o, Claude, Gemini 2) to test our conjecture that the interpretability gap is robust to LLM capability.

6. Conclusion

We described a SNOMED CT and ICD-10-aligned clinical KG, a parameterised Cypher-template retriever over it, and an expert-rated evaluation on 150 clinical questions. The system improves expert-rated accuracy and reduces unsupported claims versus a vector-only RAG baseline, and substantially improves traceability because the KG path carries SNOMED CT concept identifiers through to cited evidence. We frame this as a replication, in a SNOMED CT-aligned clinical setting, of a finding that is now consistent across several recent studies on ontology-grounded RAG. The worked example and per-category breakdown are intended to give readers a concrete sense of where the structured path helps and where it does not.

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Declaration on Generative AI

During the preparation of this work, the author used **Claude (Anthropic)** in order to: **Grammar and spelling check**, and **Paraphrase and reword** portions of the manuscript in response to reviewer feedback. The author did not use any AI tool to generate experimental results, figures, the Cypher template, or bibliographic entries. Quantitative results, the clinical question set, expert ratings, and per-category analysis are produced by the author and the human expert panel. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the publication’s content.

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